

Equivalence of Lorenz Transformation and the Pesudo Rotation

Bakground: Rotation in Classical $x - y$ Coordinates

As shown in *Fig.1*, we have two Cartesian coordinates systems of dimension 2, and their origins are the same point, denoted by O , their axes are colinear at first. Now we rotate one of the coordinates system, which is denoted by S' , through an angle θ , then we can use the rotation matrix to express the coordinates of any given point in S' , denoted by (x', y') in terms of the coordinates (x, y) in the static coordinates system, denoted by S :

$$\begin{bmatrix} x'_0 \\ y'_0 \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} \quad (1)$$

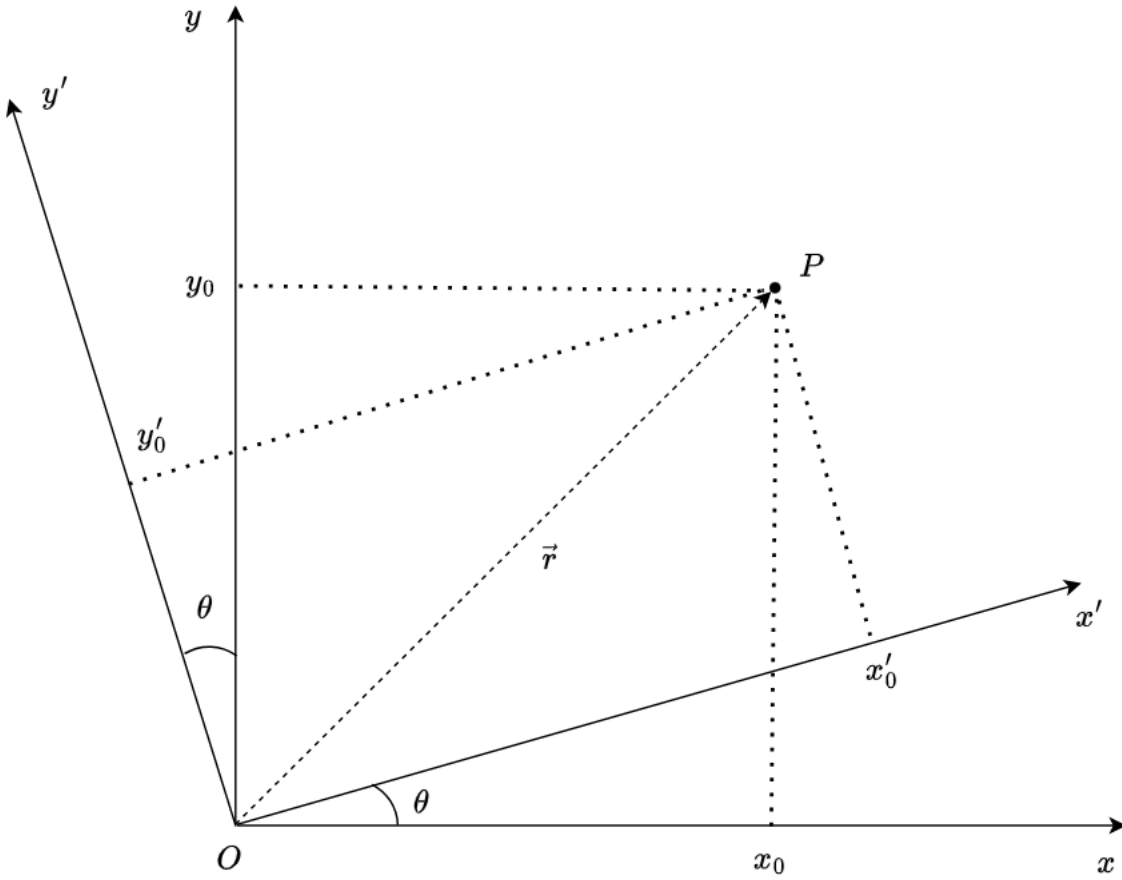


Fig.1: Rotation in Classical Coordinates

From *Equ.1* we can find that the determinant of the rotation matrix R is:

$$\det R = \cos^2 \theta + \sin^2 \theta = 1 \quad (2)$$

which conserves the invariant property of the length of the radius vector at this point $\vec{r}(x, y)$.

Analogy: Rotation in Relativistic $ct - x$ Coordinates

Like in the case of classical coordinates system, we consider the rotation of the $ct - x$ coordinates in relativity, as shown in *Fig.2*.

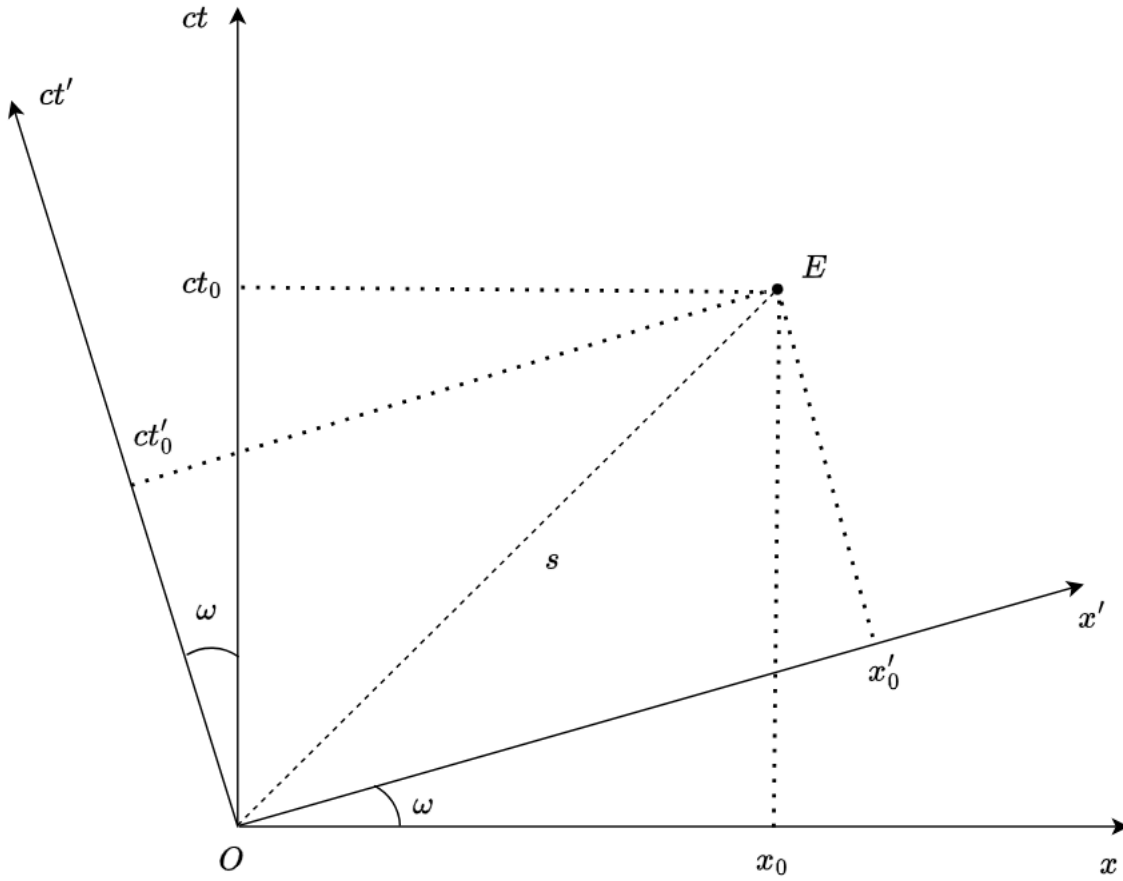


Fig.2: Pesudo Rotation in Minkowski Space

As the relativistic theory holds in Minkowski space, where Pythagorean theorem has the form:

$$x_1^2 - x_2^2 = x_3^2$$

So if we similarly want to determine an invariant quantity in the rotation, the matrix we use must be changed. The reason is that the original rotation matrix R in *Equ. 1* are combined with $\sin \theta$ and $\cos \theta$, and the property of *Equ. 2* are based on the Pythagorean theorem in Euclidean space, which is:

$$x_1^2 + x_2^2 = x_3^2$$

As a result of different space with different curvature, we lose this property, and we must find some other functions that satisfy:

$$f_1^2(x) - f_2^2(x) = 1$$

So it is easy to come up with two solutions: $\sinh x$ and $\cosh x$, also $\sec x$ and $\tan x$.

However, if we consider further, we can notice that the property:

$$\sec^2 x - \tan^2 x = 1 \quad (3)$$

can be derived from *Equ. 2*, which means it is still based on Euclidean space, so we just drop the solution, and to check the answer:

$$\cosh^2 x - \sinh^2 x = 1 \quad (4)$$

which means the pseudo rotation in Minkowski space can be formulated as:

$$\begin{bmatrix} ct'_0 \\ x'_0 \end{bmatrix} = \begin{bmatrix} \cosh \omega & -\sinh \omega \\ -\sinh \omega & \cosh \omega \end{bmatrix} \begin{bmatrix} ct_0 \\ x_0 \end{bmatrix} \quad (5)$$

where ω is the angle between these two coordinates systems.

Proof: Equivalence of the Transformation and NonEuclidean Space

We consider the rotation in the coordinates system in Minkowski space as shown in *Fig.2*, and we assume that the rotation matrix is:

$$R' = \begin{bmatrix} \cosh \omega & -\sinh \omega \\ -\sinh \omega & \cosh \omega \end{bmatrix}$$

So we expand *Equ.5*:

$$\begin{cases} ct' = \cosh \omega(ct) - \sinh \omega x \\ x' = -\sinh \omega(ct) + \cosh \omega x \end{cases}$$

As the most fundamental property of Lorentz transformation is that it is conserved for the line element ds , we compute the two line elements before and after the transformation respectively:

$$(ds)^2 = (ct)^2 - x^2 \quad (6)$$

$$(ds')^2 = (ct')^2 - x'^2 \quad (7)$$

Then we substitute *Equ.5* into *Equ.7* and obtain:

$$(ds')^2 = (\cosh \omega(ct) - \sinh \omega x)^2 - (-\sinh \omega(ct) + \cosh \omega x)^2 \quad (8)$$

By simplification, we have:

$$(ds')^2 = (\cosh^2 \omega - \sinh^2 \omega)(dx)^2 + (\sinh^2 \omega - \cosh^2 \omega)(cdt)^2 \quad (9)$$

So we have:

$$(ds')^2 = (dx)^2 - (cdt)^2 = (ds)^2 \quad (10)$$

So the rotation matrix conserves the line element in Minkowski space.

If we set: $\gamma = \cosh \omega$, $\beta = \tanh \omega$, then we obtain the new form of the rotation matrix:

$$\begin{bmatrix} \gamma & -\gamma\beta \\ -\gamma\beta & \gamma \end{bmatrix}$$

Then we substitute this new form into *Equ.5* and expand the equation:

$$x' = \gamma(-vt + x) \quad (11)$$

$$t' = \gamma(t - \frac{vx}{c^2}) \quad (12)$$

which are the formulas for time and x -coordinates of Lorentz transformation.